

Controllability and Observability of the PEM Fuel Cell System

LAGHA Khalil

GIPSA-lab — Université Grenoble Alpes

May 2026

Outline

- 1 State-Space Model
- 2 Controllability
- 3 Observability
- 4 Conclusion

Linearised model

We use the model

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$$

with states $\mathbf{x} = [P_{H_2}, m_{w,an}, m_{w,ca}]^T$ and inputs $\mathbf{u} = [i_{FC}, W_{H_2,in}]^T$.

$$A = \begin{bmatrix} -\frac{k_{an}R_{H_2}T}{V_{an}} & 0 & 0 \\ 0 & -\frac{D_w}{V_{an}t_m} & \frac{D_w}{V_{ca}t_m} \\ 0 & \frac{D_w}{V_{an}t_m} & -\frac{D_w}{V_{ca}t_m} \end{bmatrix}, \quad B = \begin{bmatrix} -\frac{M_{H_2}nR_{H_2}T}{2FV_{an}} & \frac{R_{H_2}T}{V_{an}} \\ \frac{n_dM_v}{F} & 0 \\ \frac{(n-2n_d)M_v}{2F} & 0 \end{bmatrix}$$

n : number of cells in the stack. n_d : electro-osmotic drag coefficient (water molecules per proton).

What is controllability?

A system is **controllable** if we can move the state from an initial value to a desired value using the input $\mathbf{u}(t)$.

We check the **controllability matrix**

$$\mathcal{C} = [B \quad AB \quad A^2B] \in \mathbb{R}^{3 \times 6}$$

If $\text{rank}(\mathcal{C}) = 3$, the system is controllable.

Controllability matrix

Let

$$b_1 = -\frac{M_{H_2} n R_{H_2} T}{2F V_{an}}, \quad b_2 = \frac{R_{H_2} T}{V_{an}}, \quad d_1 = \frac{n_d M_v}{F}, \quad d_2 = \frac{(n - 2n_d) M_v}{2F}$$

where n is the number of cells in the stack. so $B = \begin{bmatrix} b_1 & b_2 \\ d_1 & 0 \\ d_2 & 0 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} -\frac{k_{an} R_{H_2} T}{V_{an}} b_1 & -\frac{k_{an} R_{H_2} T}{V_{an}} b_2 \\ -\frac{D_w}{V_{an} t_m} d_1 + \frac{D_w}{V_{ca} t_m} d_2 & 0 \\ \frac{D_w}{V_{an} t_m} d_1 - \frac{D_w}{V_{ca} t_m} d_2 & 0 \end{bmatrix}$$

The 3rd row of AB is the opposite of the 2nd row.

Rank check

Since $b_2 = \frac{R_{H_2} T}{V_{an}} > 0$, P_{H_2} is directly affected by $W_{H_2,in}$.

So we only need to study the water subsystem:

$$A_w = \begin{bmatrix} -\beta & \gamma \\ \beta & -\gamma \end{bmatrix}, \quad B_w = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

Its controllability matrix is

$$C_w = [B_w \quad A_w B_w] = \begin{bmatrix} d_1 & -\beta d_1 + \gamma d_2 \\ d_2 & \beta d_1 - \gamma d_2 \end{bmatrix}$$

and

$$\det(C_w) = (d_1 + d_2)(\beta d_1 - \gamma d_2)$$

Final controllability condition

Since

$$d_1 + d_2 = \frac{nM_v}{2F} > 0$$

the rank drops only when

$$\beta d_1 = \gamma d_2$$

By replacing

$$\beta = \frac{D_w}{V_{an}t_m}, \quad \gamma = \frac{D_w}{V_{ca}t_m}, \quad d_1 = \frac{n_d M_v}{F}, \quad d_2 = \frac{(n - 2n_d)M_v}{2F}$$

we get

$$\frac{2n_d}{V_{an}} = \frac{n - 2n_d}{V_{ca}} \iff n_d = \frac{nV_{an}}{2(V_{an} + V_{ca})}$$

So

$$\text{rank}(\mathcal{C}) = 3 \iff n_d \neq \frac{nV_{an}}{2(V_{an} + V_{ca})}$$

This means the system is controllable except for that special case.

What is observability?

A system is **observable** if we can recover the state from the measured output $y = v_{fc}$.

Here

$$y = C\mathbf{x} + Du,$$

where C is the Jacobian of v_{fc} with respect to the states:

$$C = \begin{bmatrix} \frac{\partial v_{fc}}{\partial P_{H_2}} & \frac{\partial v_{fc}}{\partial m_{w,an}} & \frac{\partial v_{fc}}{\partial m_{w,ca}} \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix}$$

We test observability with

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

Full C matrix

From

$$v_{fc} = E(P_{H_2}) - v_{act}(i_{FC}) - v_{ohm}(i_{FC}, \lambda_m) - v_{con}(i_{FC})$$

only E and v_{ohm} depend on the states.

At the operating point $(\mathbf{x}_0, \mathbf{u}_0)$:

$$C = \left[\frac{4.31 \times 10^{-5} T}{P_{H_2,0}} \quad \kappa_0 \lambda'(\phi_{an,0}) \frac{R_v T}{2V_{an} p_{sat}(T)} \quad \kappa_0 \lambda'(\phi_{ca,0}) \frac{R_v T}{2V_{ca} p_{sat}(T)} \right]$$

where

$$\kappa_0 = \frac{i_0 t_m}{A_{fc} \sigma_{m,0}^2} b_{11} \exp \left[b_2 \left(\frac{1}{303} - \frac{1}{T} \right) \right]$$

and

$$\lambda'(a) = 17.81 - 79.70a + 108a^2, \quad \sigma_{m,0} = \sigma_m(\lambda_{m,0}, T)$$

If λ_m is kept constant (for example $\lambda_m = 14$), then v_{fc} does not depend on $m_{w,an}$ or $m_{w,ca}$.

In that case

$$C_{red} = \begin{bmatrix} \frac{4.31 \times 10^{-5} T}{P_{H_2,0}} & 0 & 0 \end{bmatrix}$$

- In the full model, the three terms of C can be non-zero.
- In the reduced model, only P_{H_2} appears in the output.

Controllability:

- P_{H_2} is directly affected by $W_{H_2,in}$ and also by i_{FC} .
- The water subsystem is controllable if
$$n_d \neq \frac{nV_{an}}{2(V_{an} + V_{ca})}.$$
- So the full system is controllable except in that case.

The observability result depends on the voltage model used.

Observability:

- In the full hydration-dependent voltage model, C has three non-zero terms.
- In the reduced model with fixed λ_m , the true output row is
$$C_{red} = \begin{bmatrix} \frac{4.31 \times 10^{-5} T}{P_{H_2,0}} & 0 & 0 \end{bmatrix}.$$
- So voltage alone observes P_{H_2} , but not the two water states.